

LeadCatch CRM - Project Documentation

Project Information

- **Project Name:** LeadCatch - AI-Powered CRM & Scheduling Platform
 - **Course:** CNC111 - Network and Web Programming
 - **Phase:** Phase 2 (Full-Stack Implementation)
 - **Supervisor:** Dr. Ahmed Anter
 - **Submission Date:** December 2025
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Project Overview

LeadCatch is a modern AI-powered Customer Relationship Management (CRM) platform designed for sales teams. This Phase 2 implementation is a full-stack application with real backend APIs, database integration, and AI features.

Key Features

Feature	Description
 Dashboard	Real-time metrics, analytics, and AI insights
 Lead Management	Full CRUD with AI scoring, tagging, import/export
 Appointments	Calendar, event types, public booking pages
 Communication	Messaging interface (demo)
 Companies	Company management with contact associations
 Contacts	Contact management with deduplication
 Pipeline	Kanban board for deal management
 Tasks	Task tracking with priorities and due dates
 Theme System	Dark/Light mode with persistence
 Responsive	Mobile-first design

What's New in Phase 2

- ✔ Real database with Supabase (PostgreSQL)
- ✔ Authentication system with Supabase Auth
- ✔ RESTful API routes
- ✔ AI lead scoring and enrichment
- ✔ Production-ready build (74 pages)
- ✔ TypeScript for type safety

Team Structure

Name	Role	Responsibilities
Moaz El Garawany	Tech Lead	Infrastructure, AI, Coordination
Ahmed Hossam	Frontend Lead	UI/UX, All Pages, Components
Ahmed Mahmoud	Backend Engineer	LeadCatch Module APIs
Anas Salem	Backend Engineer	Appointments Module APIs

Technologies Used

Frontend Stack

Technology	Version	Purpose
Next.js	14.2.1	React framework with App Router
React	18.3.1	UI library
TypeScript	5.x	Type safety
Tailwind CSS	3.4.1	Styling
Shadcn/ui	Latest	UI component library
Framer Motion	Latest	Animations
Lucide React	Latest	Icons

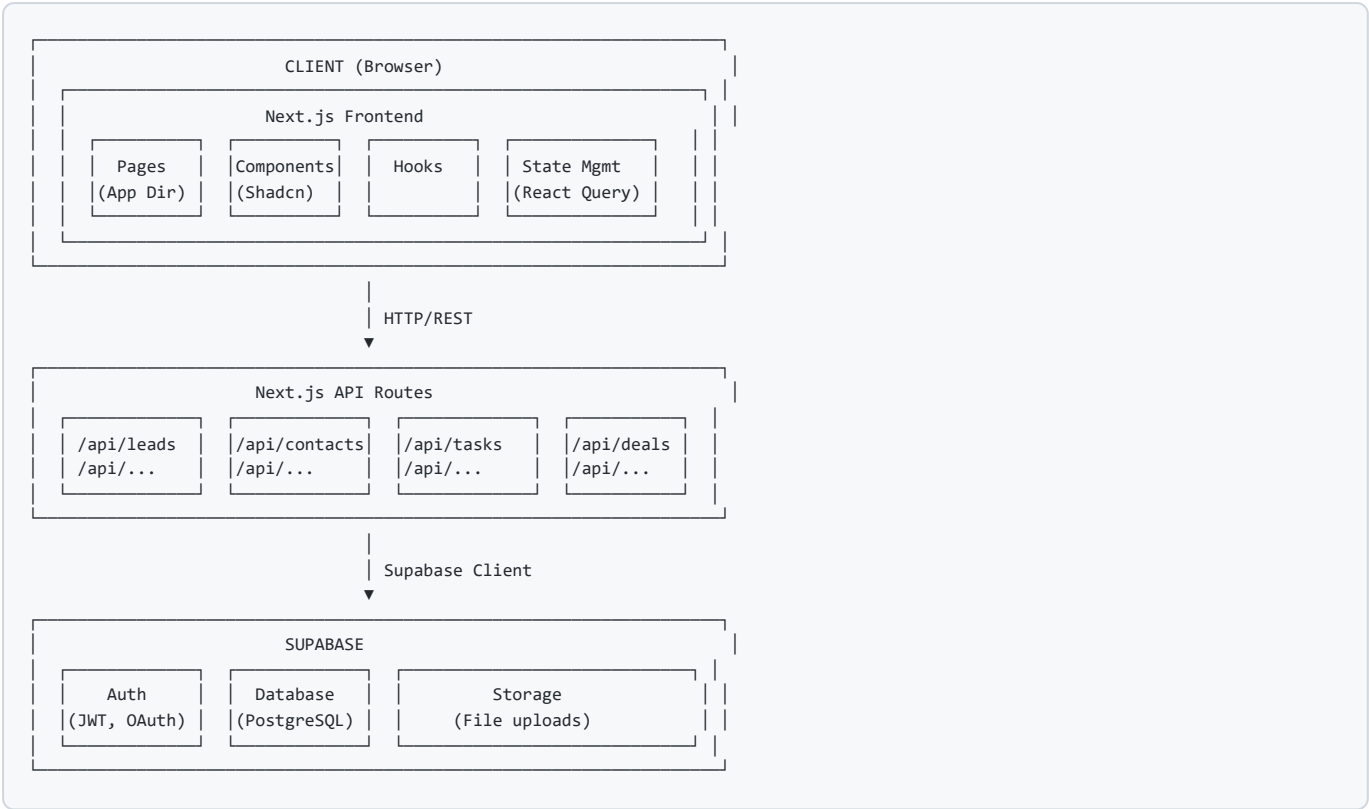
Backend Stack

Technology	Purpose
Next.js API Routes	REST API endpoints
Supabase	Database, Auth, Storage
PostgreSQL	Primary database
Drizzle ORM	Database queries

AI & Integrations

Service	Purpose
OpenAI GPT-4	Lead scoring, enrichment
Supabase Auth	Authentication
(Planned) Google Calendar	Calendar sync

System Architecture



Functional Requirements

FR1: Authentication

ID	Requirement	Status
FR1.1	User registration with email/password	✓
FR1.2	User login with credentials	✓
FR1.3	Password reset via email	✓
FR1.4	Protected routes (middleware)	✓

FR2: Lead Management

ID	Requirement	Status
FR2.1	Create, read, update, delete leads	✓
FR2.2	Filter by status/source	✓
FR2.3	Search by name/email/company	✓
FR2.4	AI lead scoring	✓
FR2.5	CSV import/export	✓
FR2.6	Tag management	✓
FR2.7	Duplicate detection	✓

FR3: Contact & Company Management

ID	Requirement	Status
FR3.1	CRUD operations for contacts	✓
FR3.2	CRUD operations for companies	✓
FR3.3	Associate contacts with companies	✓
FR3.4	Contact deduplication	✓

FR4: Pipeline & Deals

ID	Requirement	Status
FR4.1	Kanban board view	✓
FR4.2	Drag-and-drop stage changes	✓
FR4.3	Deal value tracking	✓
FR4.4	Pipeline analytics	✓

FR5: Tasks

ID	Requirement	Status
FR5.1	Create/edit/delete tasks	✓
FR5.2	Priority levels	✓
FR5.3	Due date tracking	✓
FR5.4	Task completion status	✓

FR6: Appointments

ID	Requirement	Status
FR6.1	Calendar view	✓
FR6.2	Event types management	✓
FR6.3	Availability settings	✓
FR6.4	Public booking pages	✓

FR7: Analytics

ID	Requirement	Status
FR7.1	Lead conversion metrics	✓
FR7.2	Pipeline analytics	✓
FR7.3	Activity tracking	✓

Non-Functional Requirements

NFR1: Performance

- Page load under 3 seconds
- 74 pages optimized for production
- Static generation where possible

NFR2: Security

- Row-Level Security (RLS) on database
- JWT authentication
- Input validation with Zod schemas
- Protected API routes

NFR3: Responsiveness

- Mobile-first design (320px+)
- Tablet support (768px+)
- Desktop optimization (1024px+)

NFR4: Accessibility

- Keyboard navigation
- ARIA labels
- Color contrast compliance

Database Schema

Core Tables



API Endpoints

Leads API

Method	Endpoint	Description
GET	/api/leads	List leads with filters
POST	/api/leads	Create lead
GET	/api/leads/[id]	Get lead by ID
PUT	/api/leads/[id]	Update lead
DELETE	/api/leads/[id]	Delete lead
POST	/api/leads/score	AI score lead
POST	/api/leads/import	Import CSV
GET	/api/leads/export	Export CSV

Contacts API

Method	Endpoint	Description
GET	/api/contacts	List contacts
POST	/api/contacts	Create contact
PUT	/api/contacts/[id]	Update contact
DELETE	/api/contacts/[id]	Delete contact

Companies API

Method	Endpoint	Description
GET	/api/companies	List companies
POST	/api/companies	Create company
PUT	/api/companies/[id]	Update company
DELETE	/api/companies/[id]	Delete company

Deals API

Method	Endpoint	Description
GET	/api/deals	List deals
POST	/api/deals	Create deal
PUT	/api/deals/[id]	Update deal
DELETE	/api/deals/[id]	Delete deal

Tasks API

Method	Endpoint	Description
GET	/api/tasks	List tasks
POST	/api/tasks	Create task
PUT	/api/tasks/[id]	Update task
DELETE	/api/tasks/[id]	Delete task

Appointments API

Method	Endpoint	Description
GET	/api/appointments	List appointments
POST	/api/appointments	Create appointment
GET	/api/appointments/availability	Get availability
POST	/api/appointments/book	Book appointment

File Structure

```
crm-ai/
├── app/                                # Next.js App Router
│   ├── (auth)/                        # Auth pages (login, signup)
│   ├── (dashboard)/                  # Main app pages
│   │   ├── dashboard/                # Dashboard page
│   │   ├── leads/                    # Leads management
│   │   ├── contacts/                 # Contacts management
│   │   ├── companies/                # Companies management
│   │   ├── pipeline/                 # Deals pipeline
│   │   ├── tasks/                    # Tasks management
│   │   ├── appointments/             # Appointments
│   │   ├── analytics/                # Analytics
│   │   ├── communication/            # Messaging (demo)
│   │   └── settings/                 # User settings
│   ├── (marketing)/                  # Public pages
│   ├── (booking)/                    # Public booking
│   └── api/                           # API routes
├── components/                         # React components
│   ├── ui/                           # Shadcn components
│   ├── layout/                        # Layout components
│   └── shared/                         # Shared components
├── lib/                                # Utilities
│   ├── supabase/                     # Supabase client
│   └── utils/                         # Helper functions
├── modules/                            # Business logic
├── types/                             # TypeScript types
└── public/                            # Static assets
```

Installation & Setup

Prerequisites

- Node.js 18+
- pnpm package manager
- Supabase account

Quick Start

```
# 1. Navigate to project
cd crm-ai

# 2. Install dependencies
pnpm install

# 3. Set up environment
cp .env.example .env.local
# Edit .env.local with your Supabase credentials

# 4. Run development server
pnpm dev

# 5. Build for production
pnpm build
```

Environment Variables

```
NEXT_PUBLIC_SUPABASE_URL=your_supabase_url
NEXT_PUBLIC_SUPABASE_ANON_KEY=your_anon_key
```

Stakeholders

Primary Stakeholders

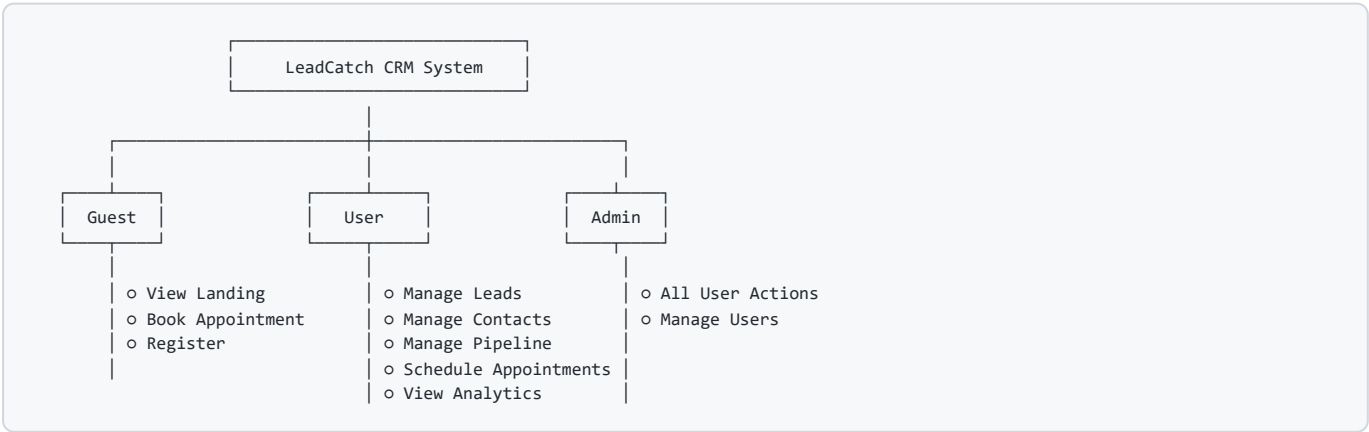
Stakeholder	Role	Interest
Sales Teams	End Users	Manage leads and appointments
Sales Managers	End Users	Monitor team performance
Marketing Teams	End Users	Track lead generation
Business Owners	Decision Makers	Increase revenue

Secondary Stakeholders

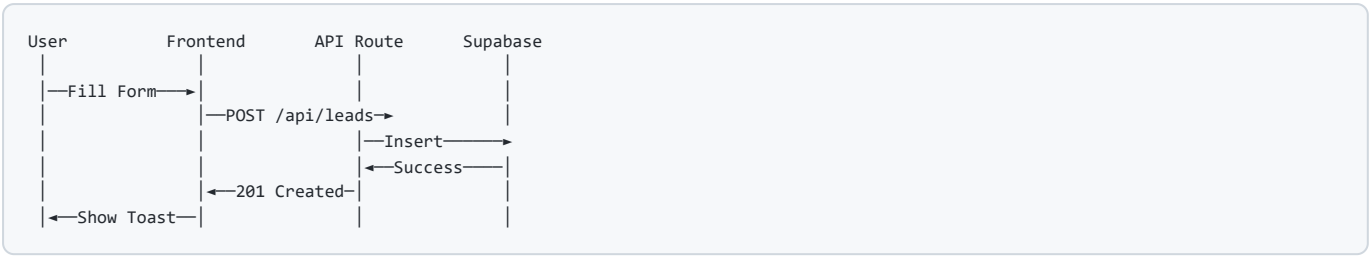
Stakeholder	Role	Interest
IT Admins	Technical Support	System maintenance
Developers	Technical Team	System development
Prospects	External Users	Easy scheduling

UML Diagrams

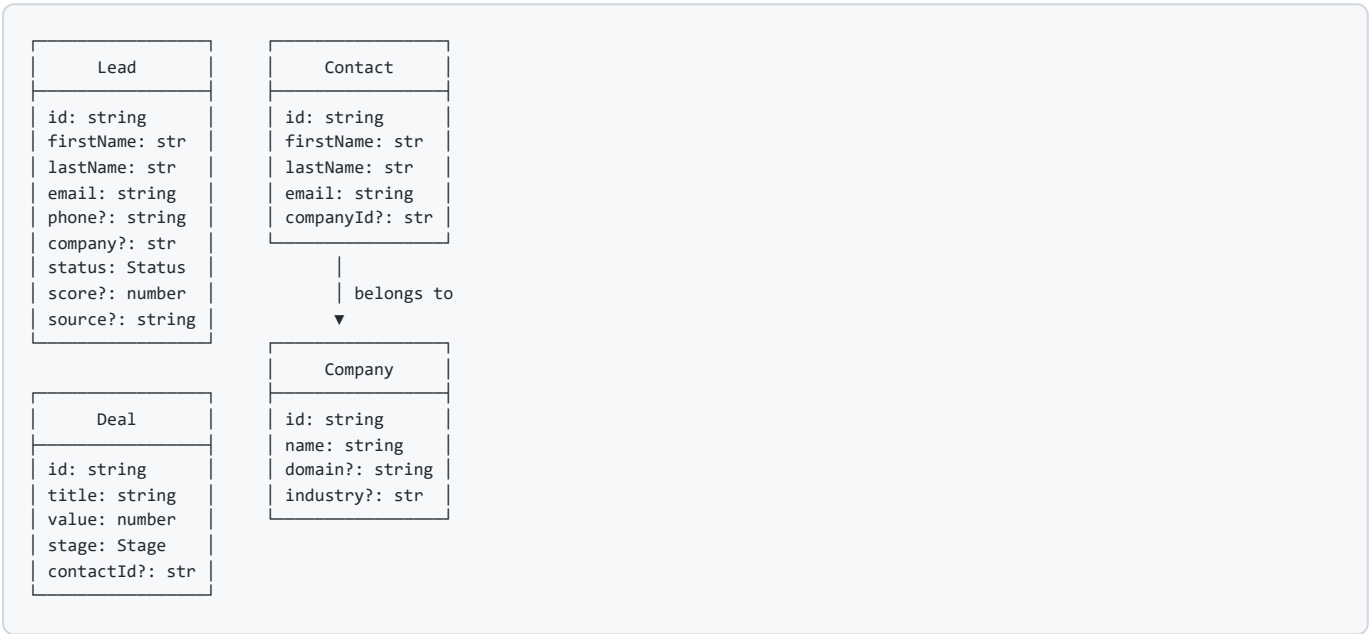
Use Case Diagram



Sequence Diagram - Lead Creation



Class Diagram (TypeScript Types)



Build Status

- ✔ 74 pages compiled successfully
- ✔ 0 errors, 5 warnings (minor)
- ✔ Production-ready build
- ✔ All API routes functional

Demo Notes

Features marked with “📄 Demo Data” in the UI use mock data for demonstration:

- Communication module
- AI Insights widget
- Settings: API Keys, Team, Billing, Integrations

All other features are fully connected to the backend API.

Credits

- **Icon Library:** Lucide React
 - **UI Components:** Shadcn/ui
 - **Database:** Supabase
 - **Deployment:** Vercel
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LeadCatch CRM - Phase 2 Full-Stack Implementation December 2025